

Autonomous Energy Routing and Redistribution Architecture for Large-Scale Battery Storage Systems with Battery Diagnostics

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Abstract—This paper presents a unified battery management module (UBMM), which achieves autonomous energy redistribution and online battery diagnosis in large-scale battery energy storage systems. All energy routing paths are coupled together through a single high-frequency capacitively coupled AC link. The equalization current is automatically regulated by the cell voltage deviation, enabling fully autonomous energy circulation among battery cell groups and across modules without complex control and any selection switches. The diagnostic circuit utilizes a switched-inductor circuit to precisely shape the charge/discharge current profile, the two battery cell groups form a closed-loop energy-circulator. The resulting voltage and current waveforms are sampled, and internal parameters are accurately identified using a modified particle swarm optimization (*m*-PSO) algorithm. The ohmic resistance extracted from these parameters is then directly mapped to state-of-health (SOH). A prototype comprising two Nissan Leaf battery modules (in total 16 cells, with a 4-series-2-parallel (4s2p) configuration per module, 3.7 V nominal per cell) was constructed to experimentally validate the proposed topology. Equalization performance was comprehensively evaluated under both single-module and dual-module operation. The battery diagnostic function was experimentally verified. Measurement results were compared with those from a Bio-Logic SP-300 potentiostat, confirming the high accuracy of the proposed diagnostic circuit.

Index Terms—Battery equalizer, battery diagnostics, battery management system, energy storage systems, lithium-ion batteries.

I. INTRODUCTION

WITH the rapidly increasing production of electric vehicles, a massive number of electric vehicles (EVs) are expected to reach their end-of-life in the foreseeable future, resulting in a substantial volume of retired lithium-ion traction batteries that require proper recycling and secondary utilization [1]. Despite retaining significant residual capacity (typically 70–80%), these retired batteries present a critical waste management challenge and a missed resource opportunity [2]. Repurposing them for second-life applications in

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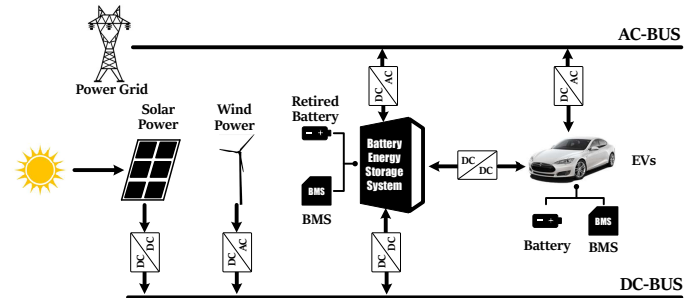


Fig. 1. Typical structure of battery-based energy storage systems.

stationary energy storage offers a sustainable and economically viable solution, as shown in Fig. 1. Among potential applications, integration with photovoltaic (PV) systems emerges as a particularly promising avenue, leveraging the batteries' capabilities to mitigate PV intermittency and provide flexible energy management [3].

Deploying second-life batteries presents significant technical challenges due to divergent cell aging during their first life, resulting in substantial cell-to-cell variations in capacity, internal resistance, and state-of-health (SOH) [4] [5]. These inherent heterogeneities inevitably cause voltage and state-of-charge (SOC) imbalances within packs, leading to diminished usable capacity, accelerated and uneven degradation, and heightened safety risks like overcharging and thermal runaway [6]. To harness their potential while mitigating risks, the battery management system (BMS) is pivotal [7]. This requires low control complexity and easy-to-implement cell equalization, precise SOC estimation, and reliable SOH quantification [8].

As illustrated in Fig. 2, existing active battery equalization topologies are primarily implemented in two forms: one employs a DC–DC converter combined with a cell-selection switch network, while the other utilizes a multi-winding transformer for energy coupling. The DC–DC converter with cell-selection switch network offers high flexibility, enabling arbitrary directed energy transfer, supports large equalization currents for fast equalization speed, and readily accommodates closed-loop control and advanced strategies [9]. However, as the number of batteries increases, the control of the switch becomes more complicated, and the increase of the battery pack voltage leads to an increase in the voltage stress of the semiconductor switch in the converter [10]. Multi-winding

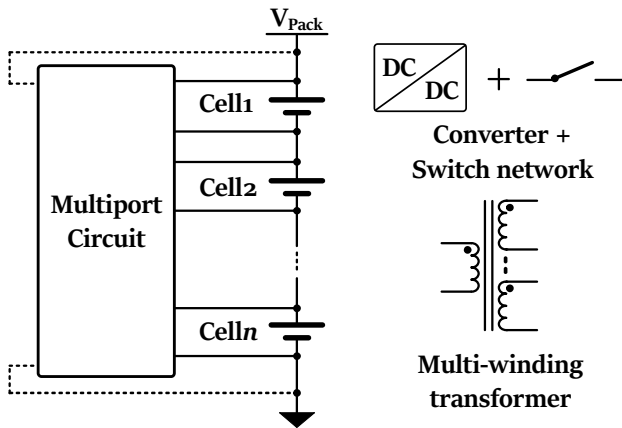


Fig. 2. Conventional battery equalization architecture.

transformer-based topologies significantly reduce active component count and eliminate the need for complicated control strategies [11]. However, the use of a large number of windings makes the realization of the transformer very challenging, and equalization performance is limited by transformer manufacturing consistency [12].

Accurate monitoring of battery states is critically important to ensure the operational safety of energy storage systems and to maximize battery life [13]. Battery state estimation methods can generally be categorized into two main approaches [14]: offline experimental methods [15] and online estimation techniques [16]. Offline methods involve testing the battery under controlled laboratory conditions, where the SOC and SOH are determined by performing full charge and discharge cycles [17]. The full charge capacity (FCC) is obtained via coulomb counting and compared with the nominal capacity to estimate SOH. In addition, predefined load profiles can be applied to analyze transient voltage responses and extract equivalent circuit model parameters, such as time constants of RC networks [18]. Other techniques, including single/double pulse tests, electrochemical impedance spectroscopy (EIS), and constant phase element (CPE) analysis, are also employed for offline SOH estimation. However, these methods are typically limited to single-cell analysis, require disconnecting the battery from the system, are time-consuming, and dissipate stored energy, making them unsuitable for online applications. In contrast, online estimation methods leverage real-time measurements of terminal voltage, current, and temperature to infer internal battery parameters, making them suitable for time-critical applications such as uninterruptible power supplies (UPS). For example, unscented Kalman filters (UKF) have been employed to estimate battery model parameters with high accuracy, although their convergence may require long data sequences and is sensitive to model uncertainty and initial conditions. Similarly, optimization-based methods such as sequential quadratic programming (SQP) can be used for parameter identification, but they also depend heavily on extended data records [19]. While online approaches offer advantages in adaptability and real-time monitoring, they face challenges in terms of computational complexity and

robustness under varying operating conditions [20].

To address the dual demands of energy equalization and condition monitoring in large-scale battery storage systems, this paper proposes a unified battery management module (UBMM) that integrates both battery equalization and real-time SOC/SOH estimation. By leveraging an autonomous energy routing circuit and a second-order battery model, the system enables energy-efficient inter-battery transfer while simultaneously extracting intrinsic parameters through a modified particle swarm optimization (m -PSO) algorithm. This integrated approach not only ensures energy equalization but also provides accurate and non-intrusive health diagnostics, offering a compact and scalable solution for next-generation battery management systems.

Compared with [9], this paper contributes the following:

- (1) a multi-module PWM synchronization scheme enabling scalability to 48 UBMM units;
- (2) integration of an embedded diagnostic circuit with online parameter extraction.

The remainder of this paper is structured as follows. Sec. II describes the proposed circuit configuration and its operating principles. Sec. III presents the operating modes of the battery diagnostic circuit. Sec. IV gives the battery parameter extraction algorithm. Sec. V provides experimental equalization and diagnostic results of two Nissan Leaf battery modules. Sec. VI compares the proposed architecture with state-of-the-art topologies. Sec. VII discusses the system constraints and limitations. Finally, Sec. VIII gives the conclusion.

II. PROPOSED SYSTEM AND OPERATING PRINCIPLES

A. Battery Energy Storage System Structure

Fig. 3(a) illustrates the energy storage system consisting of M battery packs (BP), each comprising N battery modules (BM). The entire battery system can be connected to a bidirectional DC/DC or DC/AC converter, depending on the application and load. As shown in Fig. 3(b), each battery module comprises eight individual battery cells. These cells are configured into four series-connected groups, with each group containing two cells connected in parallel. Power management is implemented using UBMM units, where each UBMM interfaces with two adjacent groups.

The nomenclature is defined as follows.

Battery Cell: The fundamental physical unit (a single battery).

Battery Group: Two distinct cells connected in parallel.

UBMM: The dedicated hardware unit that executes active equalization and embedded diagnostics. Specifically, one UBMM connects to and manages exactly two battery groups.

Module: The primary building block comprising four battery groups connected in series. Each Module requires two UBMMs.

4s2p Configuration: 2 cells are first connected in parallel to form a battery group, and 4 such groups are then connected in series.

B. Proposed UBMM Architecture

Fig. 4 shows the structure of the proposed UBMM for a single battery pack. The module consists of two functional

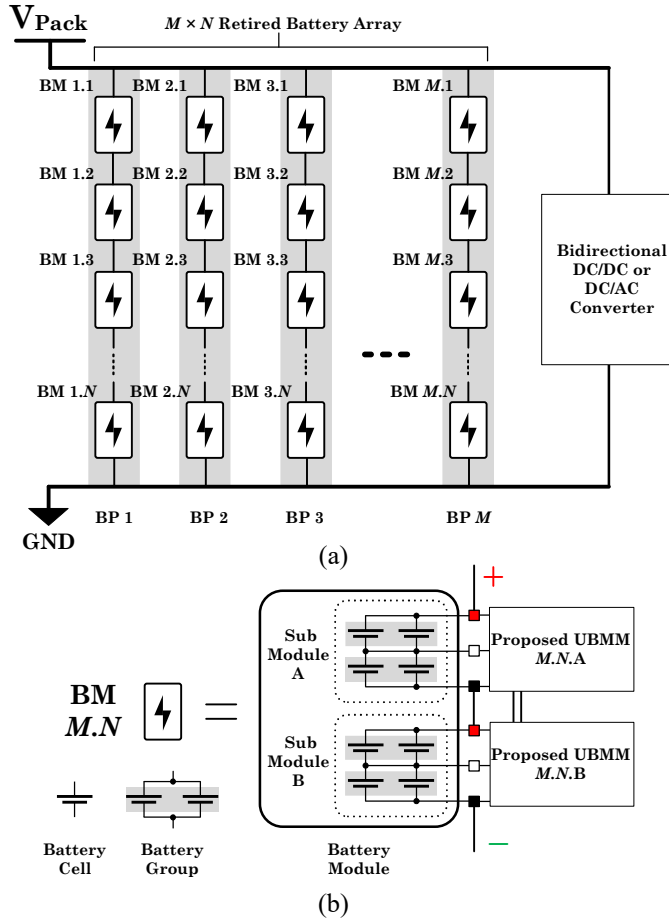


Fig. 3. (a) Battery energy storage system with multi-pack architecture. It comprises a cluster of $M \times N$ retired battery modules. (b) Battery module internal structure and UBMM connection.

units: the battery equalizer and the battery diagnostic circuit. The equalizer is based on ZETA-derived structure, and all equalizers are connected together via the ac link. Each battery cell group is capable of transferring energy to or receiving energy from other groups, either within the same module or across different modules. The battery diagnostic circuit is based on a buck-boost converter that serves as an active diagnostic source, with the two battery groups serving as the input and output, respectively. Although the buck-boost circuit has been used for active battery equalization, it only achieves equalization between two battery units connected to its two ports. Due to its bidirectional power flow capability, this circuit is employed for battery diagnosis by generating the required excitation currents.

C. Operating Principles of the Equalizer

As shown in Fig. 5(a), the proposed equalizer achieves energy redistribution among series-connected battery groups through an AC-coupled architecture. A high-frequency AC link is established using a single-switch converter and decoupling capacitors. Multiple cell-specific rectifier circuits are connected in an input-parallel, output-series (IPOS) configuration, each coupled to an individual battery group via capacitors

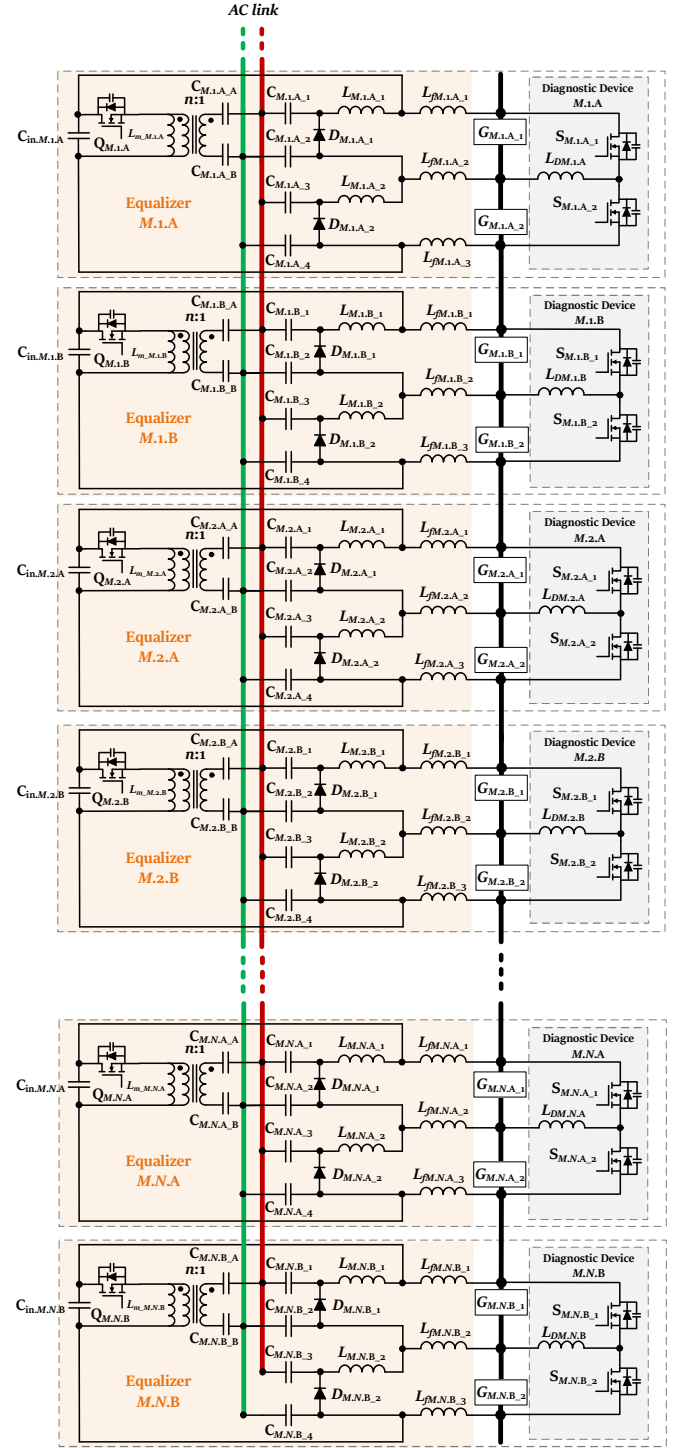


Fig. 4. Schematic of the proposed unified battery management module (UBMM).

($C_{M.1.A.1} \sim C_{M.N.A.4}$). This structure enables autonomous charging and discharging of battery cell groups based on their voltage differences, without requiring selector switches or centralized control. The key waveforms in the steady state during equalization is shown in Fig. 5(b).

The entire battery module provides the input power, drawing a common average current of $i_{M.1.A}$. In this case, the group contributes a discharging current equal to the $i_{M.1.A}$. In con-

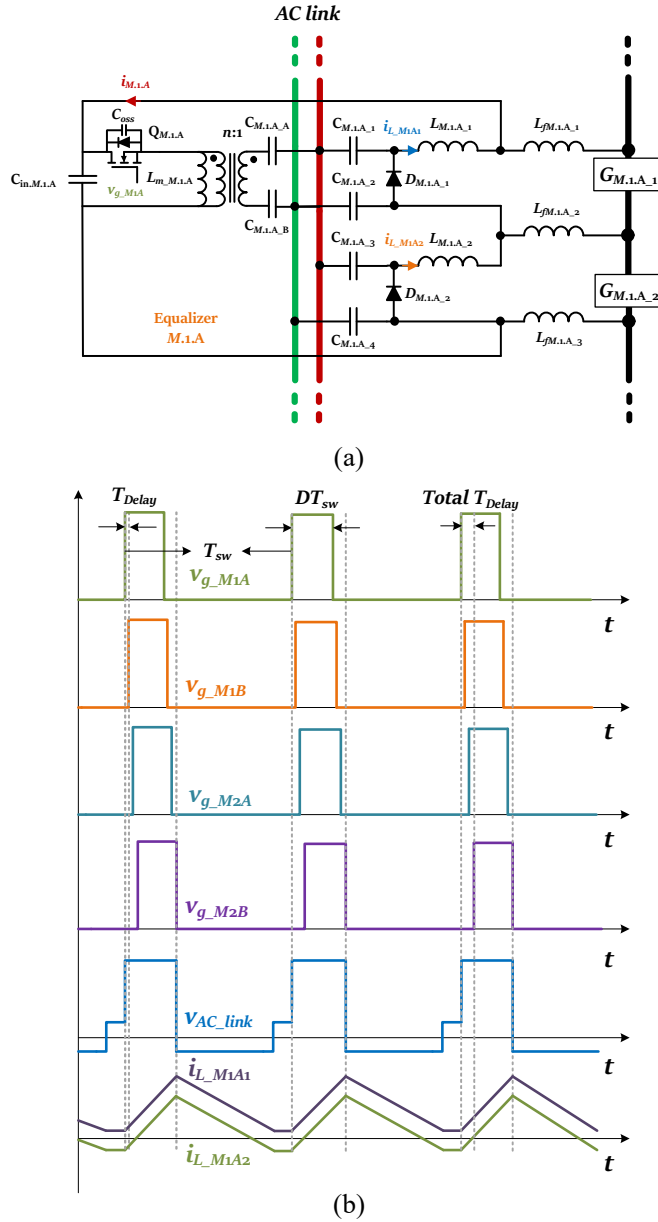


Fig. 5. Topological operation of the proposed equalization circuit. (b) Key waveforms in the steady state during equalization.

trast, the group with the lowest voltage receives charge. The equalization process proceeds in a fully autonomous manner, with the equalization current naturally decreasing as the group voltage differences diminish. It can be assumed that the voltage of $G_{M,1,A,1}$ is higher than that of $G_{M,1,A,2}$, the equalization current of low-voltage groups is shown as

$$i_{low} = i_{L,M,1,A,2} - i_{M,1,A}. \quad (1)$$

For high voltage groups, $i_{L,M,1,A,1} = 0$, the equalization current is shown as

$$i_{high} = i_{M,1,A}. \quad (2)$$

When voltage convergence is achieved, the equalization current approaches zero, ensuring self-termination of the process without external control.

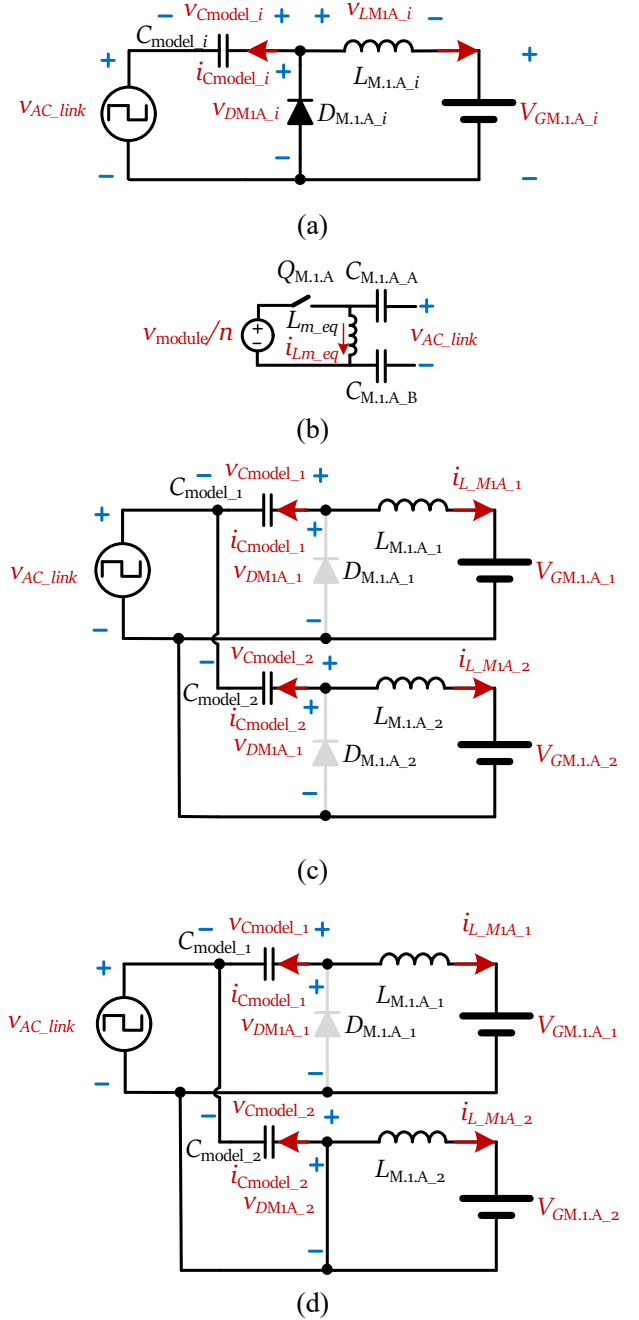


Fig. 6. Equivalent circuits. (a) Simplified circuit of the converter. (b) Simplified circuit of dc/ac converter. (c) Equivalent circuit of mode I and III, (d) Equivalent circuit of mode II.

D. Equalization Circuit Analysis

For the sake of simplicity, the simplified equivalent circuit is depicted in Fig. 6, where all components are assumed ideal. The signal delay among modules is neglected in the analysis. Fig. 7 shows the current waveforms for analysis. The equivalent circuit for each battery branch is depicted in Fig. 6(a). In this model, the equivalent coupling capacitance

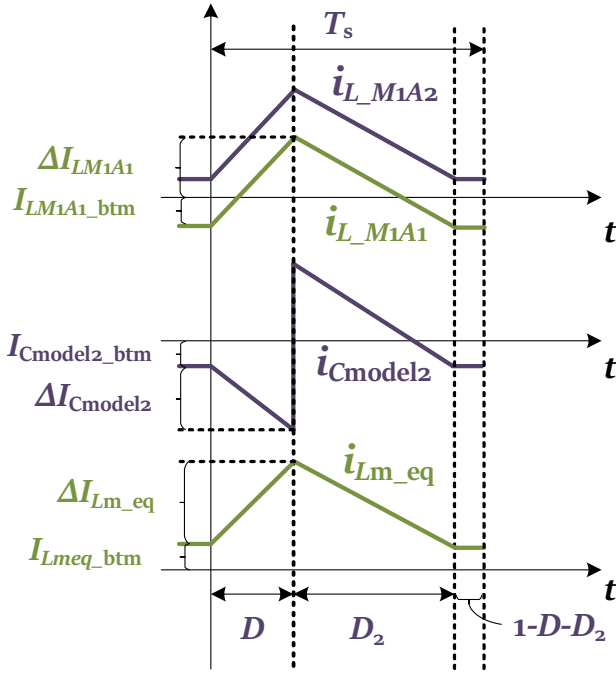


Fig. 7. Current Waveforms of $L_{M.1.A.1}$, $L_{M.1.A.2}$, C_{model2} , and L_{meq} .

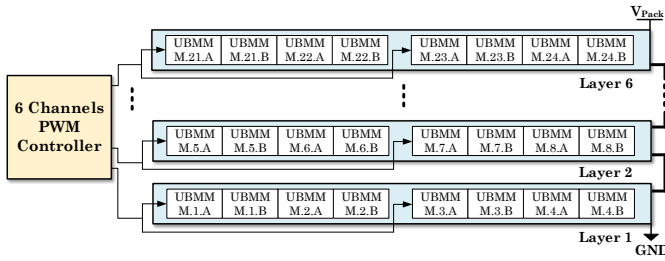


Fig. 8. Diagram of the PWM synchronous structure.

is represented by two capacitors connected in series. Thus

$$C_{model-i} = \frac{C_{M.1.A.1}C_{M.1.A.2}}{C_{M.1.A.1} + C_{M.1.A.2}} \quad (3)$$

Based on the Kirchhoff's Voltage Law (KVL),

$$v_{AC_link} + v_{Cmodel-i} - v_{LM1A-i} - V_{GM.1.A-i} = 0 \quad (4)$$

where $v_{AC_link} >_{T_s} 0$, and $< v_{LM1A-i} >_{T_s} 0$. Therefore,

$$< v_{Cmodel-i} >_{T_s} = V_{GM.1.A-i} \quad (5)$$

Fig. 6(b) shows the simplified equivalent circuit of the equalizer. The equivalent magnetizing inductance $L_{m_eq} = L_{m_M.1.A}/n^2$.

Mode-I [$0 - DT_s$, Fig. 6(c)]

The primary main switch $Q_{M.1.A}$ is turned ON. The input voltage V_{in} is applied across the primary winding, causing the magnetizing current i_{Lm_eq} to increase linearly. The secondary AC-link voltage is established as $v_{AC_link} = V_{module}/n$. Consequently, the secondary diodes $D_{M.1.A.1}$ and $D_{M.1.A.2}$ are reverse-biased. Moreover, the capacitor current $i_{Cmodel2}(t)$ is equal to the inductor

current $i_{L_{M1A2}}(t)$. The accumulated charge $Q_{Cmodel2}$ is given by

$$Q_{Cmodel2-I} = - \left(I_{Cmodel2_btm} + \frac{1}{2} \Delta I_{Cmodel2} \right) DT_s \quad (6)$$

Mode-II [$DT_s - (D + D_2)T_s$, Fig. 6(d)]

The main switch $Q_{M.1.A}$ is turned OFF, initiating the demagnetization of the transformer. The AC-link voltage v_{AC_link} reverses polarity to maintain the magnetic flux. Due to the voltage differential between the battery groups ($V_{GM.1.A.2} < V_{GM.1.A.1}$), diode $D_{M.1.A.2}$ is preferentially forward-biased, clamping the AC-link voltage to $v_{AC_link} = -V_{GM.1.A.2}$. Meanwhile, $D_{M.1.A.1}$ remains reverse-biased. The voltage across the diode $D_{M.1.A.1}$ in the higher-voltage branch ($G_{M.1.A.1}$) is evaluated as

$$\begin{aligned} v_{DM1A1} &= v_{AC_link} + V_{DM1A1} \\ &= -V_{GM.1.A.2} + V_{GM.1.A.1} > 0 \end{aligned} \quad (7)$$

Thus, diode $D_{M.1.A.1}$ is reliably reverse-biased and remains in the off-state throughout the entire switching cycle. During this interval, i_{Lm_eq} releases its stored energy to the lowest-voltage branch. During this phase, applying Kirchhoff's Current Law (KCL) at the secondary node reveals that C_{model2} absorbs both the reflected magnetizing current and the current from the non-charging high-voltage branch, yielding $i_{Cmodel2}(t) = i_{Lm_eq}(t) + i_{L_{M1A1}}(t)$. The corresponding charge $Q_{Cmodel2}$ is:

$$\begin{aligned} Q_{Cmodel2-II} &= \left[\left(I_{Lmeq_btm} + \frac{1}{2} \Delta I_{Lm_eq} \right) \right. \\ &\quad \left. + \left(I_{LM1A1_btm} + \frac{1}{2} \Delta I_{LM1A1} \right) \right] D_2 T_s \end{aligned} \quad (8)$$

Mode-III [$(D + D_2)T_s - T_s$, Fig. 6(c)]

This state commences when the current flowing through the clamping diode $D_{M.1.A.2}$ naturally decays to zero, causing it to turn OFF. Thus, the charge $Q_{Cmodel2-III}$ is

$$Q_{Cmodel2-III} = -I_{Cmodel2_btm}(1 - D - D_2)T_s \quad (9)$$

In steady state, based on charge balance of coupling capacitor C_{model2} ,

$$Q_{Cmodel2-I} + Q_{Cmodel2-II} + Q_{Cmodel2-III} = 0 \quad (10)$$

Based on volt-second balance of equivalent magnetizing inductance L_{m_eq} ,

$$\frac{V_{module}}{n} \cdot DT_s - V_{GM.1.A.2} \cdot D_2 T_s = 0 \quad (11)$$

Thus, D_2 is determined by the low voltage battery group in the module and is derived as

$$D_2 = \frac{V_{module} D}{n V_{GM.1.A.2}} \quad (12)$$

By using (6), (8), (9), (10), and (12), the average current of $L_{M.1.A.2}$ can be shown to be

$$I_{L_{M1A2}} = \frac{V_{module}^2 D^2}{2 V_{GM.1.A.2} f_s} \left(\frac{1}{L_{m_M.1.A}} + \frac{2}{n^2 L} \right) \quad (13)$$

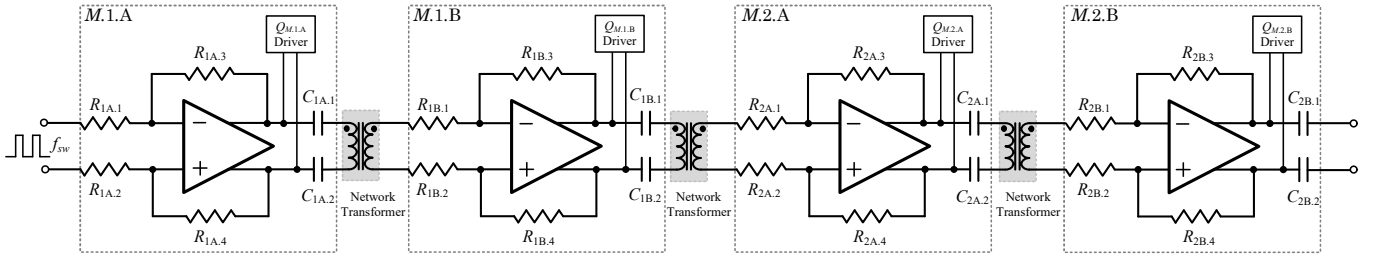


Fig. 9. Cascaded PWM synchronization architecture based on differential amplifiers.

where $L = L_{M.1.A.1} = L_{M.1.A.2}$.

Since $D_{M.1.A.1}$ is persistently OFF, the inductor $L_{M.1.A.1}$ and capacitor C_{model_1} form a strict series branch. Based on the principle of capacitor charge balance, the average current through C_{model_1} is zero. Therefore, $i_{L_{M.1.A.1}} = 0$, the higher-voltage battery group receives zero net equalization current. During equalization, the output power equals the input power, therefore,

$$V_{module} I_{M.1.A} = V_{GM.1.A.2} I_{L_{M.1.A.2}} \quad (14)$$

$$I_{M.1.A} = \frac{V_{module} D^2}{2f_s} \left(\frac{1}{L_{m_{M.1.A}}} + \frac{2}{n^2 L} \right) \quad (15)$$

E. PWM Signal Synchronization

As illustrated in Fig. 4, all UBMMs are interconnected via the AC link. To guarantee that the AC link voltages across all UBMM modules remain phase-aligned, it is necessary to enforce synchronization of the PWM signals within each module. Therefore, the PWM distribution architecture shown in Fig. 8 is designed. A single six channels synchronous PWM controller generates 6 phase-locked PWM carriers. Each carrier is fed to one layer of the cascaded stack, where it is split 1-to-2 via a splitter. The resulting twelve synchronized PWM signals are then transmitted through Ethernet cables to the respective UBMMs. As shown in Fig. 9, the reference PWM signal is first injected into the initial module, where it is reshaped and amplified by the differential amplifier and then coupled to the next module through the network transformer; the same process is repeated stage by stage until the last module. Since all stages share identical amplifier structures and the propagation delay introduced by the network transformer is negligible, the timing skew between adjacent modules is essentially determined only by the fixed propagation delay of the differential amplifier. Consequently, the PWM edges observed by all modules exhibit a small and well-controlled relative delay, which satisfies the timing requirements of the multi-module synchronized control.

F. Operating Principles of the Diagnostic Circuit

As shown in Fig. 10(a), the second-order RC model represents the battery using an open-circuit voltage source OCV , a series ohmic resistance R_0 , and two parallel RC branches (R_1, C_1) and (R_2, C_2) . The resistor R_0 models the instantaneous voltage drop due to internal resistance, while the two RC branches capture different polarization dynamics:

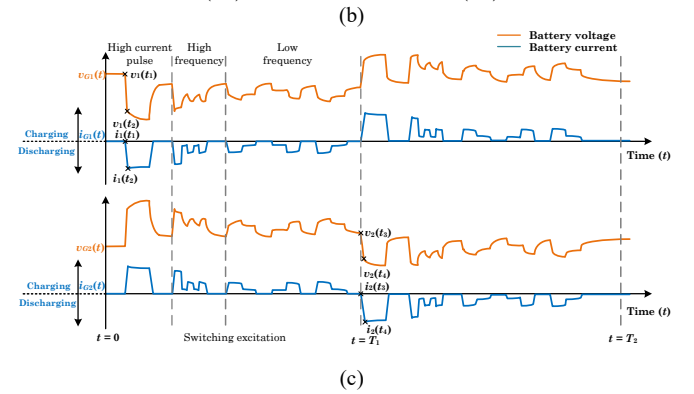
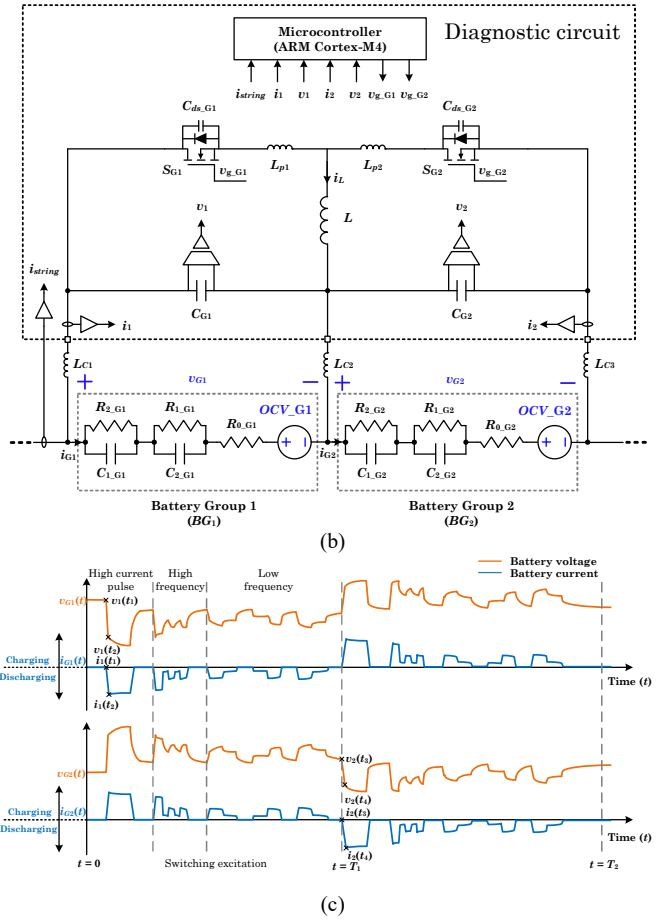
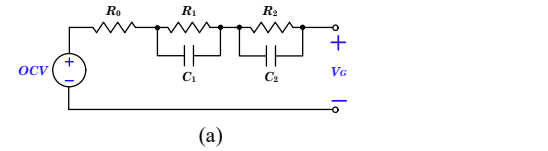


Fig. 10. (a) Simplified Thevenin 2-order battery model. (b) Circuit structure of the proposed diagnostic circuit. (c) Voltage and current profiles of the two battery groups.

(R_1, C_1) reflects the short-term electrochemical polarization, and (R_2, C_2) accounts for slower diffusion effects. This model achieves a good balance between accuracy and computational efficiency, and is widely used in real-time SOC and SOH estimation within BMS [21].

The proposed method is based on firstly sampling the voltage across and current through the two batteries or battery

sets connected to the diagnostic circuit and then applying a m -PSO to extract the intrinsic parameters of the two batteries or battery sets to determine their SOC and SOH. As illustrated in Fig. 10(b), the power conversion stage comprises an inductor L , two MOSFETs (S_{G1} and S_{G2}), and two capacitors (C_{G1} and C_{G2}). The voltages (v_{G1} , v_{G2}) and currents (i_{G1} , i_{G2}) of the two battery groups are derived from the sampled terminal voltages (v_1 , v_2) and currents (i_1 , i_2) of the diagnostic circuit, along with the main string current i_{string} . This circuit configuration facilitates controlled energy transfer between battery groups BG_1 and BG_2 . The diagnostic circuit and the two battery groups form an individual diagnostic unit in a large-scale system. Each unit does not affect the normal operations of the others.

Fig. 10(c), shows the low-frequency profiles of i_1 and i_2 , and the corresponding voltage waveforms, v_1 and v_2 . The diagnostic current profile is adapted from [22]. There are three excitation stages, including 1) high current pulse excitation stage, 2) high-frequency excitation stage, and 3) low-frequency excitation stage. During the time interval $[0, T_1]$, BG_1 is discharging and BG_2 is charging. During the time interval $[T_1, T_2]$, BG_1 is charging and BG_2 is discharging.

The high pulse current excitation stage is used to calculate the approximated values of $R_{0,G1}$ and $R_{0,G2}$. Upon a sudden change of the battery currents, the reactance of the equivalent capacitances $C_{1,G1}$, $C_{2,G1}$, $C_{1,G2}$, and $C_{2,G2}$ are small and can be considered to be short-circuited. Thus, the estimated values of $R_{0,G1}$ and $R_{0,G2}$ are calculated by the changes of the battery voltage and current, which is shown as

$$R_{0,G1,est} = \frac{v_1(t_1) - v_1(t_2)}{i_1(t_1) - i_1(t_2)} \quad (16)$$

$$R_{0,G2,est} = \frac{v_2(t_3) - v_2(t_4)}{i_2(t_3) - i_2(t_4)} \quad (17)$$

$R_{0,G1,est}$ and $R_{0,G2,est}$ are used to define the search boundaries for determining the values of $R_{0,G1}$ and $R_{0,G2}$ in the m -PSO algorithm.

G. Switching Frequency Selection

The buck-boost converter in the diagnostic architecture is designed to operate in the discontinuous current mode (DCM). This operational mode ensures that the inductor current naturally drops to zero before the end of each switching period.

Based on the volt-second balance and the geometric relationship of the triangular current pulses, a fundamental design trade-off exists among the inductance L , the switching frequency f_{sw} , and the maximum required average testing current I_{max} . The detailed mathematical derivation is provided in Appendix. For a given inductor L , peak current I_{L-pk} , and battery voltages v_{G1} , v_{G2} , the switching frequency must satisfy

$$f_{sw} \leq \frac{v_{G1}v_{G2}^2}{2I_{max}L(v_{G1} + v_{G2})^2} \quad (18)$$

Equation (18) establishes the quantitative relationship among the maximum excitation current, inductance, and switching frequency.

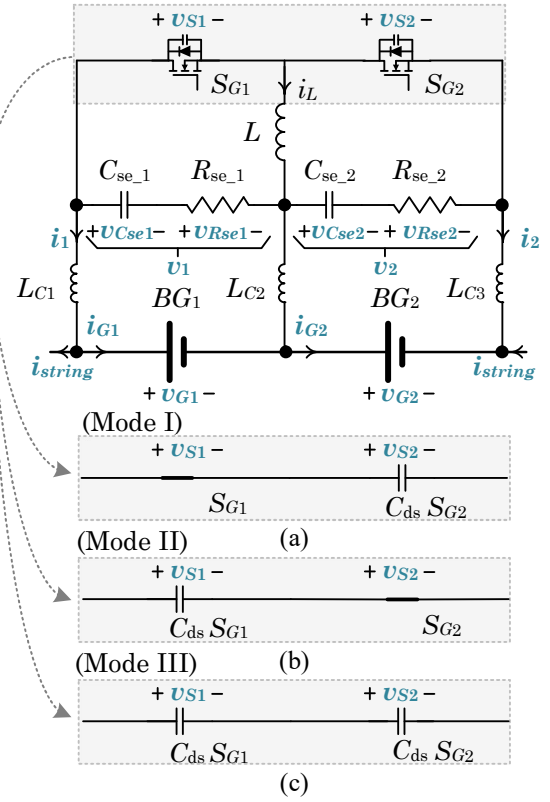


Fig. 11. Operating modes of the diagnostic circuit.

III. OPERATING MODE

A. Analysis Assumptions

Each device has three operating modes – Modes I, II, and III, shown in Fig. 11. The switching period is $T_{sw} = 1/f_{sw}$, where f_{sw} is the switching frequency.

Ideally, $v_1 = v_{G1}$, $v_2 = v_{G2}$, $i_1 = i_{G1} + i_{string}$, respectively. However, the cable inductances, denoted by $L_{c1} - L_{c3}$, cause battery current oscillations with the output capacitances of the MOSFETs upon switching, thus introducing discrepancies in obtaining actual battery response and extracting intrinsic battery parameters. The capacitors C_{G1} and C_{G2} are used to provide low impedance current paths for mitigating such issue.

For simplicity, the following assumptions have been made in the following analysis:

- 1) To facilitate installation of the device, the connection cables are of the same length. Thus, their inductances are assumed to be the same and are equal to L_c . The resistances of the cable are assumed to be the same and are equal to R_c .
- 2) C_{se1} and C_{se2} have the same value and are equal to C . The equivalent series resistance (R_{se1} and R_{se2}) of C_{se1} and C_{se2} are the same and are equal to R_c .
- 3) The practical value of L_{p1} and L_{p2} are less than 50nH. Its impedance is significantly smaller than the impedance of its associated circuit loop in Fig. 11, implying that its voltage can be neglected in the loop.

4) Under a high-frequency switching operation, the reactances of C_{1_G1} , C_{2_G1} , C_{1_G2} , and C_{2_G2} in Fig. 10(b) are very small.

5) S_{G1} and S_{G2} have zero on-state resistance, but have their junction capacitances, C_{ds_G1} and C_{ds_G2} , the same and are equal to C_{ds} .

B. Operation Mode

Mode I - Fig.11(a) [$t_A - t_B$]

During this subinterval, S_1 is turned on. The BG_1 discharges energy to L . Based on Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), the following can be derived.

$$i_L(t) = \frac{1}{L} \int_{t_A}^{t_B} v_1(\tau) d\tau. \quad (19)$$

$$v_1(t) = v_{G1}(t) + 2L_c \frac{di_{G1}(t)}{dt} - L_c \frac{di_{G2}(t)}{dt}. \quad (20)$$

$$v_2(t) = v_{G2}(t) - L_c \frac{di_{G1}(t)}{dt} + 2L_c \frac{di_{G2}(t)}{dt}. \quad (21)$$

$$i_{G1}(t) = -i_L(t) - C \frac{d[v_1(t) - v_{Rse1}(t)]}{dt} - C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (22)$$

$$i_{G2}(t) = -C \frac{d[v_2(t) - v_{Rse2}(t)]}{dt} - C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (23)$$

Mode II - Fig.11(b) [$t_B - t_C$]

When S_2 conducts, L is discharged. Thus,

$$i_L(t) = \frac{1}{L} \int_{t_B}^{t_C} v_2(\tau) d\tau + i_L(t_B). \quad (24)$$

$$v_1(t) = v_{G1}(t) + 2L_c \frac{di_{G1}(t)}{dt} - L_c \frac{di_{G2}(t)}{dt} - v_{R1}(t). \quad (25)$$

$$v_2(t) = v_{G2}(t) - L_c \frac{di_{G1}(t)}{dt} + 2L_c \frac{di_{G2}(t)}{dt} - v_{R2}(t). \quad (26)$$

$$i_{G1}(t) = -C \frac{d[v_1(t) - v_{Rse1}(t)]}{dt} - C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (27)$$

$$i_{G2}(t) = i_L(t) - C \frac{d[v_2(t) - v_{Rse2}(t)]}{dt} - C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (28)$$

Mode III - Fig.11(c) [$t_C - t_D$] This mode starts at t_C when S_1 and S_2 are off. Thus,

$$i_L(t) = C_{ds} \frac{dv_{S1}(t)}{dt} - C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (29)$$

$$v_1(t) = v_{G1}(t) + 2L_c \frac{di_{G1}(t)}{dt} - L_c \frac{di_{G2}(t)}{dt} - v_{R1}(t). \quad (30)$$

$$v_2(t) = v_{G2}(t) - L_c \frac{di_{G1}(t)}{dt} + 2L_c \frac{di_{G2}(t)}{dt} - v_{R2}(t). \quad (31)$$

$$i_{G1}(t) = -C \frac{d[v_1(t) - v_{Rse1}(t)]}{dt} + C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (32)$$

$$i_{G2}(t) = -C \frac{d[v_2(t) - v_{Rse2}(t)]}{dt} + C_{ds} \frac{dv_{S2}(t)}{dt}. \quad (33)$$

A PI controller is employed to regulate the magnitude and direction of i_{G1} and i_{G2} by alternately modulating the duty cycles of the two active switches, which controls the duration of Mode I. The voltages v_1 and v_2 , and the currents i_1 and i_2 , are sampled to estimate the intrinsic parameters of the batteries.

IV. PARAMETERS EXTRACTION ALGORITHM

The mechanism of extracting the intrinsic parameters of the batteries are given in Fig.12. The voltages, v_1 and v_2 , currents, i_1 and i_2 are sampled over a test period T with the sampling period T_{samp} . The number of sampling points over T , N , is

$$N = \frac{T}{T_{\text{samp}}}. \quad (34)$$

As the sampling frequency is significantly lower than the dynamics due to L_c , the voltage drop across L_c can be neglected. Then, it can be assumed that $v_{G1} = v_1$ and $v_{G2} = v_2$.

The battery parameters are extracted using a modified particle swarm optimization (m -PSO) engine which mitigates the local optima entrapment issue inherent in conventional PSO while accelerating the search process [22]. The population size is P . The algorithm initializes a population of particles.

$$S_i^p = \{ \text{SOC}_i^p(0), R_{0,i}^p, R_{1,i}^p, R_{2,i}^p, \tau_{1,i}^p, \tau_{2,i}^p, v_{C1,i}^p(0), v_{C2,i}^p(0) \}, i = 1, 2 \quad (35)$$

where $p = 1, 2, \dots, P$, $\text{SOC}_i^p(0)$ is candidate for the initial values of the SOC of BG_1 and BG_2 , $\tau_{1,i}^p$ and $\tau_{2,i}^p$ are time constant of RC network, and $v_{C1,i}^p(0)$ and $v_{C2,i}^p(0)$ are the initial voltages of capacitors. The battery voltage estimator depicted in Fig. 12 utilizes the measured current series, initial internal parameters, and initial open-circuit voltage $v_{oc,1}(0)$ and $v_{oc,2}(0)$ as inputs to estimate the terminal voltage. The estimation is performed according to

$$v_{Gi,\text{est}}^p[n] = v_{oc,i}(\text{SOC}_i^p[n]) + i_{Gi}[n] R_{0,i}^p + \sum_{j=1}^2 v_{Cj,i}^p[n], \quad i = 1, 2 \quad (36)$$

The open-circuit voltage $v_{oc,i}(\text{SOC}_i^p[n])$ is derived from the V_{oc} -SOC characteristics, obtained experimentally via the relaxation method in [23]. Fig. 13 presents an example for a Nissan leaf Li-ion battery.

The m -PSO generates new particles by minimizing the differences between actual and estimated battery voltages over the period T . For each particle, the fitness is evaluated using

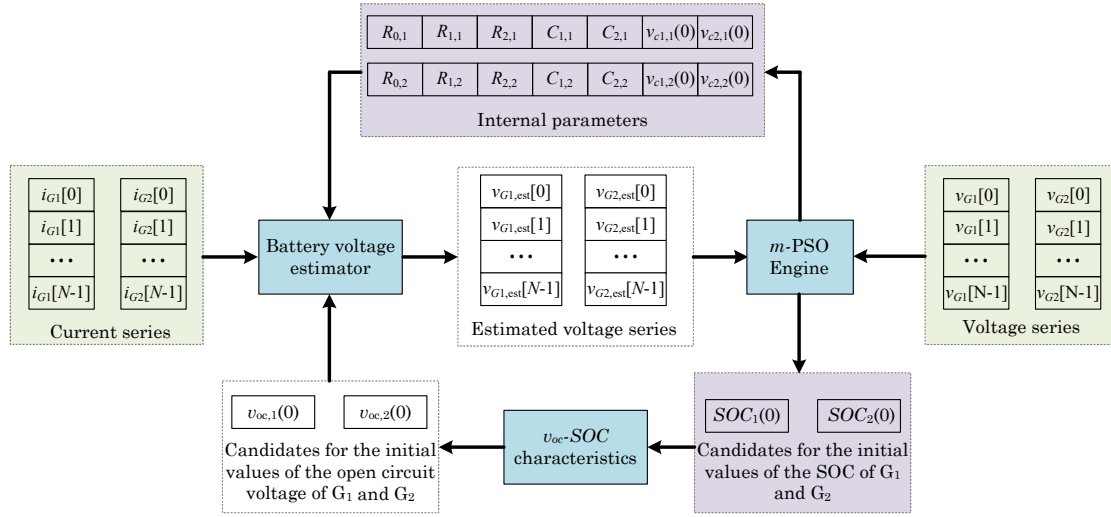


Fig. 12. Mechanism of extracting intrinsic parameters.

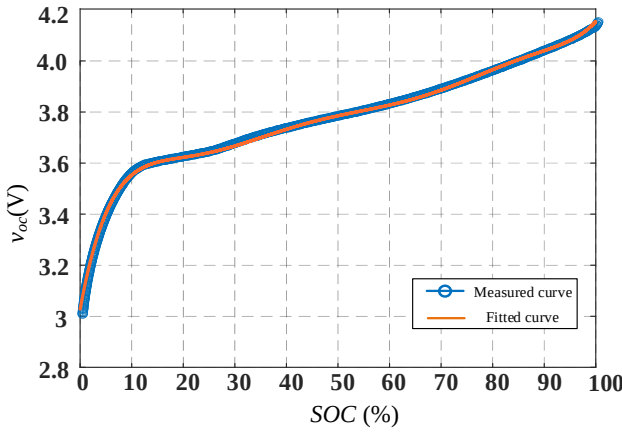


Fig. 13. $v_{oc} - SOC$ curve measured by 10A current of Nissan leaf battery.

the mean absolute error (MAE), which quantifies the average deviation between these voltage profiles and is defined as

$$MAE(S_i^p) = \frac{1}{N} \sum_{n=0}^{N-1} |v_{Gi}[n] - v_{Gi,est}^p[n]|, i = 1, 2 \quad (37)$$

The optimal parameter sets S_i^p is selected from the population as those yielding the minimum values of $MAE(S_i^p)$.

The m -PSO extraction operation is outlined as follows:

1) The batteries are excited via the charging and discharging processes depicted in Fig. 10. The voltages v_1 and v_2 , currents i_1 and i_2 , and string current i_{string} are sampled.

2) N particles or parameter sets S_i^p are initialized randomly.

3) The battery voltages of BG_1 and BG_2 are estimated using (36) based on the time series of actual battery currents and intrinsic parameters. The MAEs for all particles are evaluated via (37).

4) The velocities $V_i^{(g)}$ of all particles in the current gener-

ation g are updated as

$$v_i^{(g)} = w^{(g-1)} v_i^{(g-1)} + 2\alpha_i^{(g-1)} [S_g - S_i^{(g-1)}] + 2\beta_i^{(g-1)} [S_{H,i}^{(g-1)} - S_i^{(g-1)}], i = 1, 2 \quad (38)$$

where $V_i^{(g-1)}$ is the velocity of the each particle in the previous generation, S_g is the global best position found so far in the m -PSO process, $S_i^{(g-1)}$ is the position of the each particle in generation $g-1$, $S_{H,i}^{(g-1)}$ is the personal best position of the each particle up to generation $g-1$, $w^{(g-1)} \in (0, 1)$ is the inertia weight, $\alpha_i^{(g-1)}, \beta_i^{(g-1)} \sim U(0, 1)$ are independent random numbers.

The positions of all particles are then updated using their previous positions and velocities:

$$S_i^{(g)} = S_i^{(g-1)} + v_i^{(g)} \quad (39)$$

5) A random perturbation $\Delta S^{(g)}$ is added to the globally best particle $S_G^{(g)}$ in the current generation to yield a new candidate $\tilde{S}_B^{(g)}$, facilitating escape from local minima in the m -PSO, according to

$$\tilde{S}_B^{(g)} = S_G^{(g)} + \Delta S^{(g)} \quad (40)$$

The perturbation vector $\Delta S^{(g)}$ consists of individual perturbations for each identified parameter element χ (e.g., SOC, R_0 , R_1 , etc.). To ensure appropriate scaling, each parameter is perturbed by a magnitude $\Delta\chi^{(g)}$ bounded by its physical search range $[\chi_{min}, \chi_{max}]$:

$$\Delta\chi^{(g)} = [\chi_{max} - \chi_{min}] \times \text{Random}(0, 0.8) \quad (41)$$

where $\text{Random}(0, 0.8)$ denotes a normally distributed random number with a zero mean and a variance of 0.8. The variance dictates the exploratory behavior of the swarm. While a large variance increases the probability of escaping local minima, an excessively large value causes adverse scattering. Consistent with discussions in [24] and verified through empirical testing, a variance of 0.8 optimally balances the convergence rate and

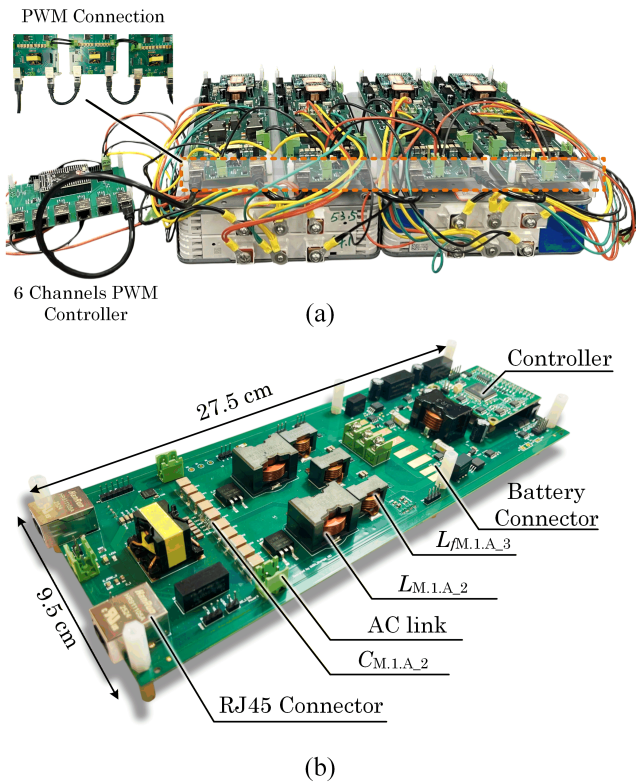


Fig. 14. (a) Photographs of the experiment setup. (b) Designed UBMM prototype.

the ability to avoid local optima without affecting identification accuracy.

Steps 3)-6) are repeated for a predefined number of generations.

V. EXPERIMENTAL VERIFICATIONS

A. Hardware Implementation

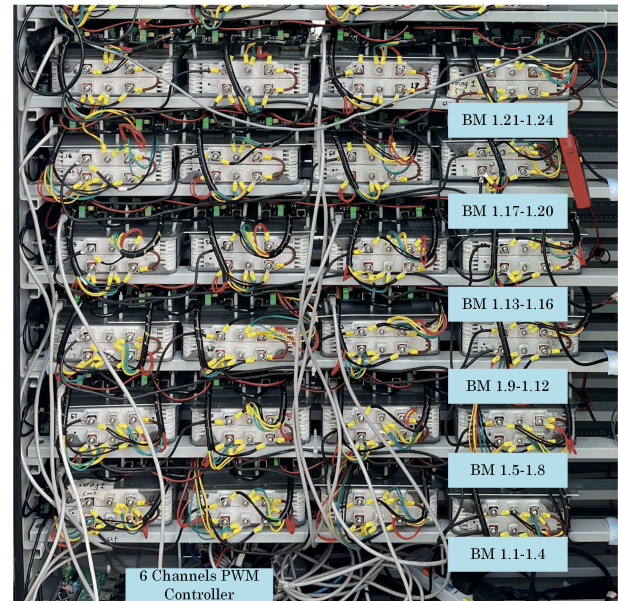
Fig. 14(a) shows a laboratory prototype with two BMs and four UBMMs. The experimental setup consists of eight series-connected retired Nissan leaf lithium-ion batteries (3.7V, 100Ah). The UBMM is shown in 14(b). The diagnostic is built around an ARM Cortex-M4 microcontroller and integrates voltage and current measurements. In Fig. 15, the design of UBMM is applied to a battery pack composed of 24 modules (a total of 192 cells / 96 groups). The pack is organized into six layers, each containing four modules, with two UBMMs serving each module. The circuit parameters are listed in Table I.

B. Battery Equalization Verification

Fig. 16 shows the key waveforms of the equalization function. Fig. 16(a) shows the measured v_{g_M1A} and v_{ds_M1A} waveforms of switch Q_{M1A} , AC-link voltage v_{AC_link} , and inductor voltage (v_{L_M1A1}) at the 200kHz switching frequency, 0.3 duty cycle, and 1:1 turns ratio. Fig. 17(a) shows the PWM signals from M.1.A to M.2.B. Fig. 17(b) show that the total propagation delay between M.1.A (first board) and M.2.B (last board) is 30ns.



(a)



(b)

Fig. 15. (a) The proposed UBMM is applicable to the deployment of large-scale energy storage systems. (b) Photograph of the assembled maximum 400 V battery pack comprising six layers with a total of 24 Nissan Leaf battery modules.

Fig. 18 presents the experimental terminal voltages of individual battery groups with fixed switching frequency of 200 kHz under different operating scenarios. The terminal voltages of all groups were measured and recorded using a voltmeter every hour throughout the entire equalization process. Fig. 18(a) shows the equalization performance when one cell group has a significantly lower voltage compared to the average voltage of the battery pack. The initial maximum voltage difference is 396 mV. The batteries were subjected to

TABLE I
CRITICAL DESIGN PARAMETERS

Component Type	Parameters
MOSFET($Q_{M.1.A}$)	BSC320N20N
Battery	Nissan Leaf
Diode($D_{M.1.A}$)	STPS30L30CG-TR
Transformer turns ratio (n)	1 : 1
Magnetizing inductance ($L_{m_M.1.A}$)	200 μ H
Equalizer switching frequency (f_{sw})	200 kHz
Decoupling capacitor ($C_{M.1.A_A}/C_{M.1.A_B}$)	13 μ F
Coupling capacitor ($C_{11} \sim C_{N2}$)	10 μ F
Inductor (L_k)	10 μ H
Filter inductor (L_{fk})	4.7 μ H
Microcontroller	STM32F407
Diagnostic switching frequency	38 kHz
Diagnostic sampling frequency	700 kHz
Diagnostic inductor (L)	3.3 μ H
Diagnostic capacitor (C_G)	270 μ F
Diagnostic MOSFET (S_G)	BSC046N10NS3

repeated charge–discharge cycles with a constant current of 10 A to emulate practical operating conditions and evaluate the performance of the active equalization system during both charging and discharging phases. As evidenced by the measured voltage trajectories, after 11 hours of continuous equalization operation, the voltage discrepancy among the four cell groups was effectively reduced to 30 mV. Fig. 18(b) shows the equalization result for eight cell groups with the initial voltages of 3.703, 3.708, 3.270, 3.667, 3.505, 3.508, 3.509, and 3.511 V, respectively. During the equalization process, the charging current is 3A and the discharging current is 5A. Due to the more randomized initial voltage distribution—unlike the typical scenario where three higher-voltage groups actively balance a single lower-voltage cell group (see Fig. 18(a))—the energy transfer paths become less directed and more diversified. Consequently, the equalization current is distributed across multiple groups simultaneously, resulting in a reduced effective equalization current for any individual cell group and, accordingly, a slower convergence rate of the overall equalization process. After 63 hours of equalization, the voltage difference is reduced to 45 mV. As shown in Fig. 19(a), the conversion efficiency of the equalization circuit consistently exceeds 85% across the target battery voltage range, achieving a peak efficiency of 86%. Fig. 19(b) presents the measured equalization efficiency curves under different initial group voltage differences. It can be observed that as the voltage discrepancy decreases, the transferred equalization power is correspondingly reduced, which leads to a gradual decline in equalization efficiency.

C. Battery Diagnosis Verification

To validate the performance of the proposed battery diagnostic circuit, the batteries were tested at an ambient temperature of 20°C. The sampling frequency is 700kHz and

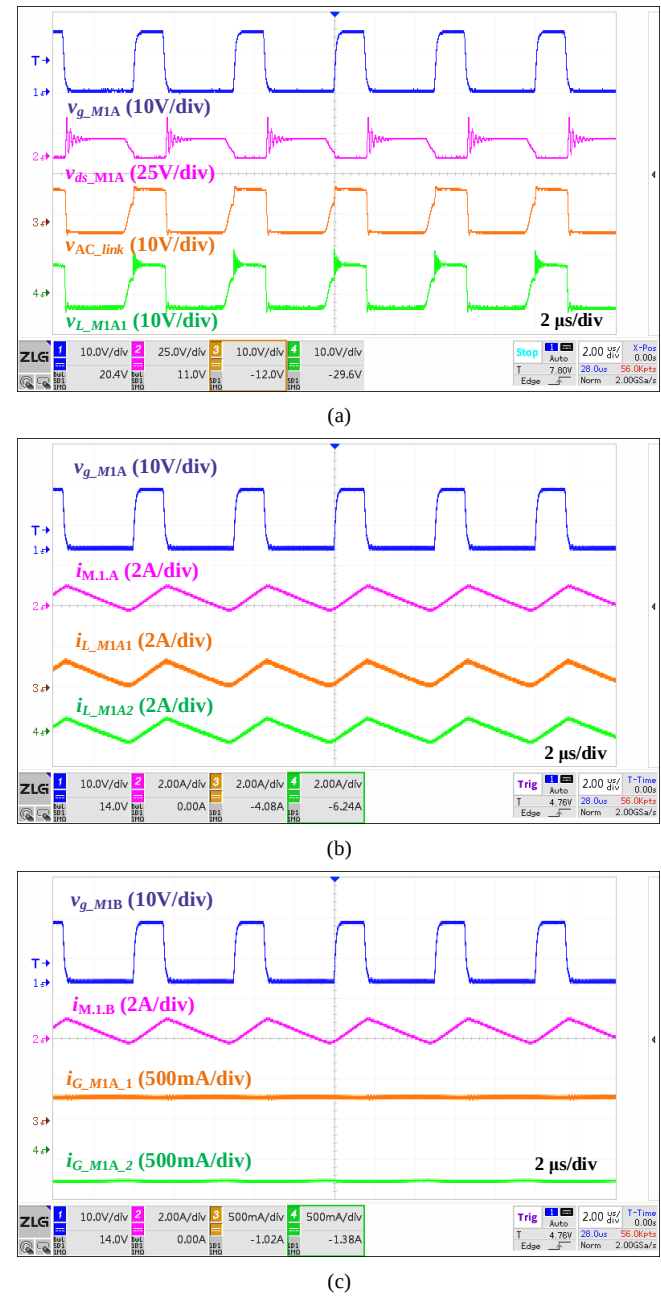
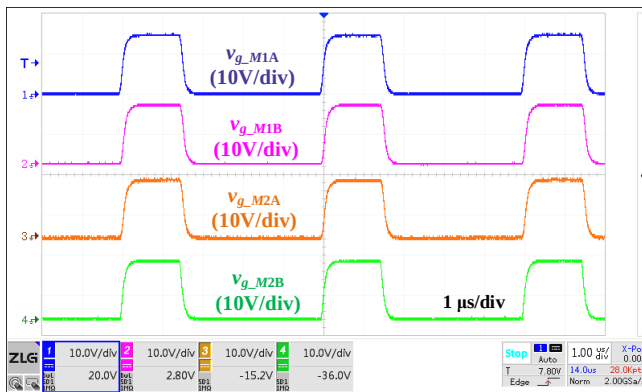


Fig. 16. Key waveforms of the equalizer. (a) Switches $Q_{M.1.A}$ gate drive signal v_{g_M1A} , drain source voltage v_{ds_M1A} , AC link voltage v_{AC_link} , and inductor voltage v_{L_M1A1} . (b) Switches $Q_{M.1.A}$ gate drive signal v_{g_M1A} , module input current i_{M1A} , and inductor current i_{L_M1A1} and i_{L_M1A2} . (c) Switches $Q_{M.1.B}$ gate drive signal v_{g_M1B} , module input current i_{M1B} , equalization net current of low-voltage battery group $i_{G_M1A_1}$, and equalization net current of high-voltage battery group $i_{G_M1A_2}$.

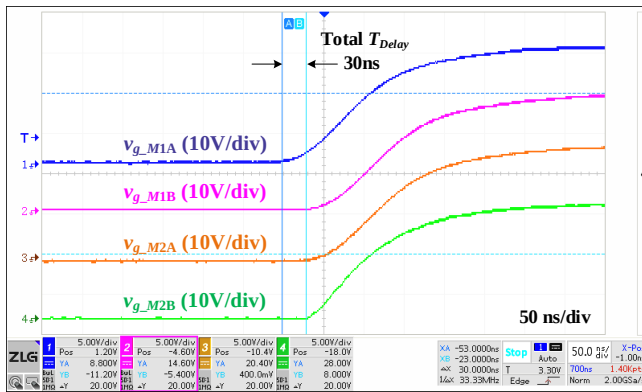
the data is down sampled to 140Hz. The m -PSO algorithm is configured with 20 particles and a maximum of 1000 generations. To guarantee convergence and avoid unreasonable solutions, the search bounds are constrained using the internal resistance limitations defined in equations (16)–(17). The total computation time to process the dataset and extract the parameters is 215 seconds. The measurement results obtained from the diagnostic circuit were compared with those acquired from a reference-grade Bio-Logic SP-300 potentiostat/galvanostat,

TABLE II
ESTIMATED INTERNAL PARAMETERS

Parameters	M _{1A} -G1	M _{1A} -G2	M _{1B} -G1	M _{1B} -G2	M _{2A} -G1	M _{2A} -G2	M _{2B} -G1	M _{2B} -G2
v_{oc} (V)	3.555	3.559	3.562	3.517	3.491	3.514	3.493	3.465
SOC (%)	22.81	23.60	24.70	12.95	10.55	12.17	10.40	8.44
SOH (%)	61.34	62.78	61.4	64.64	59.9	64.1	68.15	72
R_0 (m Ω)	2.61	2.37	2.6	2.06	2.85	2.15	1.91	1.8
R_1 (m Ω)	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49
R_2 (m Ω)	0.16	0.58	0.99	0.40	0.93	0.75	0.57	0.99
τ_1 (ms)	6.8	3.1	3.3	6.6	3.1	3.1	3.1	3.1
τ_2 (s)	2.31	3.36	0.5	0.5	4	4.83	4.85	4.81
v_{c1} (mV)	11	-33	8	-17	-14	17	16	-20
v_{c2} (mV)	-2	0	-1	0	-1	-1	-1	0



(a)



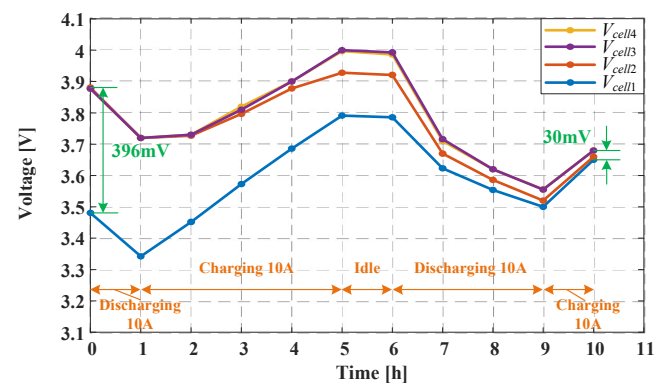
(b)

Fig. 17. (a) Switches $Q_{M.1.A}$ gate drive signal $v_{g,M1A}$, switches $Q_{M.1.B}$ gate drive signal $v_{g,M1B}$, switches $Q_{M.2.A}$ gate drive signal $v_{g,M2A}$, and switches $Q_{M.2.B}$ gate drive signal $v_{g,M2B}$. (b) The delay of the four modules PWM is 30ns.

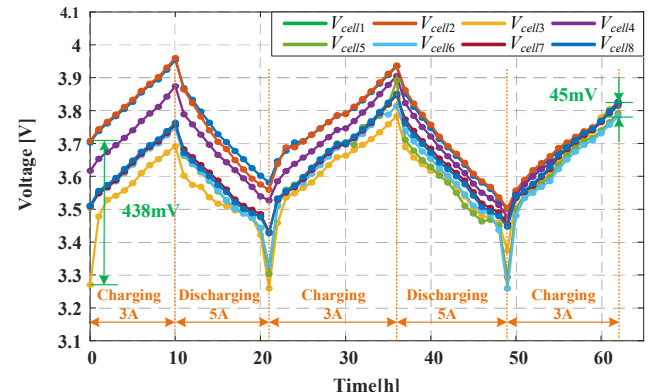
which serves as the benchmark for accuracy verification.

Fig. 20(a) shows the measured current of the battery cell group during an 18 s pulsed experiment. Fig. 20(b) provides an expanded view of the first pulse from Fig. 20(a) with a finer horizontal resolution (200 ms/div) and a vertical scale of 2 A/div.

Table II shows the parameters estimated by UBMM. Table III compares the internal resistance measurements of the proposed UBMM and the reference SP-300, and lists both



(a)



(b)

Fig. 18. Experimental results of the equalization operation with different voltage distributions. (a) Equalization results for four cell groups with an initial voltage difference of 396mV. (b) Equalization results for eight cell groups with an initial voltage difference of 438mV.

absolute and relative errors. Only cell M_{1A}-G1 reaches the maximum absolute error of 0.66 mΩ (33.85% relative error). This error is caused by random sampling noise and can be reduced by averaging multiple measurements.

Fig. 21 illustrates the relationship between the internal resistance and SOH for a Nissan Leaf battery module. The SOH is defined as the ratio of the present full-charge capacity to the rated capacity of the fresh battery. The linear equations

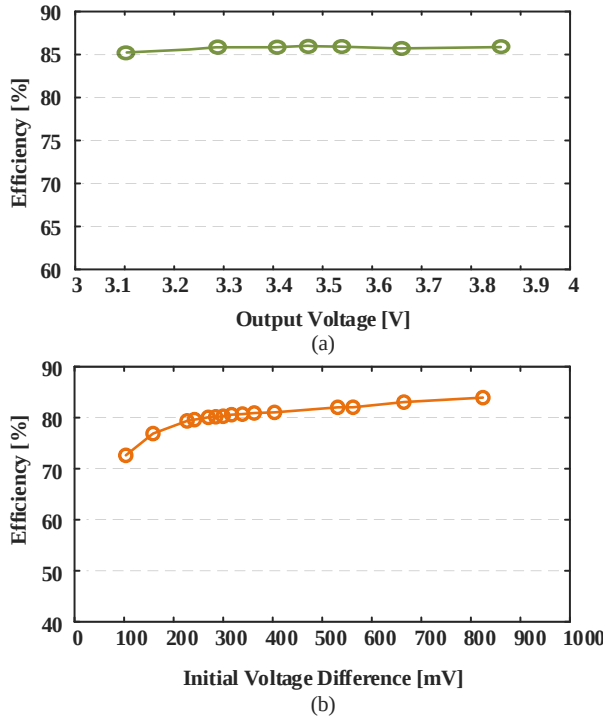


Fig. 19. (a) Measured conversion efficiency against output voltage. (b) Measured equalization efficiency against initial battery voltage difference.

TABLE III
COMPARISON OF INTERNAL RESISTANCE MEASURED BY THE UBMM AND SP-300

Cell Group	UBMM	SP-300	Absolute Error	Relative Error
M_{1A-G1}	2.61 m Ω	1.95 m Ω	0.66 m Ω	33.85%
M_{1A-G2}	2.37 m Ω	2.85 m Ω	-0.48 m Ω	-16.84%
M_{1B-G1}	2.60 m Ω	2.75 m Ω	-0.15 m Ω	-5.45%
M_{1B-G2}	2.06 m Ω	2.02 m Ω	0.04 m Ω	1.98%
M_{2A-G1}	2.85 m Ω	2.95 m Ω	-0.10 m Ω	-3.39%
M_{2A-G2}	2.15 m Ω	2.05 m Ω	0.10 m Ω	4.88%
M_{2B-G1}	1.91 m Ω	1.84 m Ω	0.07 m Ω	3.80%
M_{2B-G2}	1.80 m Ω	1.82 m Ω	-0.02 m Ω	-1.10%

depicted in this figure are piecewise linear approximations derived from the empirical non-linear aging data in [25]. As shown, the SOH degradation of the Nissan Leaf battery module exhibits two distinct phases. In the first life, when the internal resistance lies between 1 m Ω and 2 m Ω , the SOH decreases rapidly with increasing resistance. In the second life, as the internal resistance rises from 2 m Ω to 7 m Ω , the rate of SOH decline significantly slows down. The internal resistances of all cell groups G_1 and G_2 , as measured by the proposed battery diagnostic circuit, are 2.61 m Ω , 2.37 m Ω , 2.60 m Ω , 2.06 m Ω , 2.85 m Ω , 2.15 m Ω , 1.91 m Ω , and 1.80 m Ω , respectively. The estimated SOH values for the tested groups range from 59.9% to 72.0%. Fig. 22 compares the SOH values estimated by the proposed UBMM with those inferred from the internal resistance measurements obtained via the SP-300 potentiostat. The maximum SOH estimation error is 5.41%.

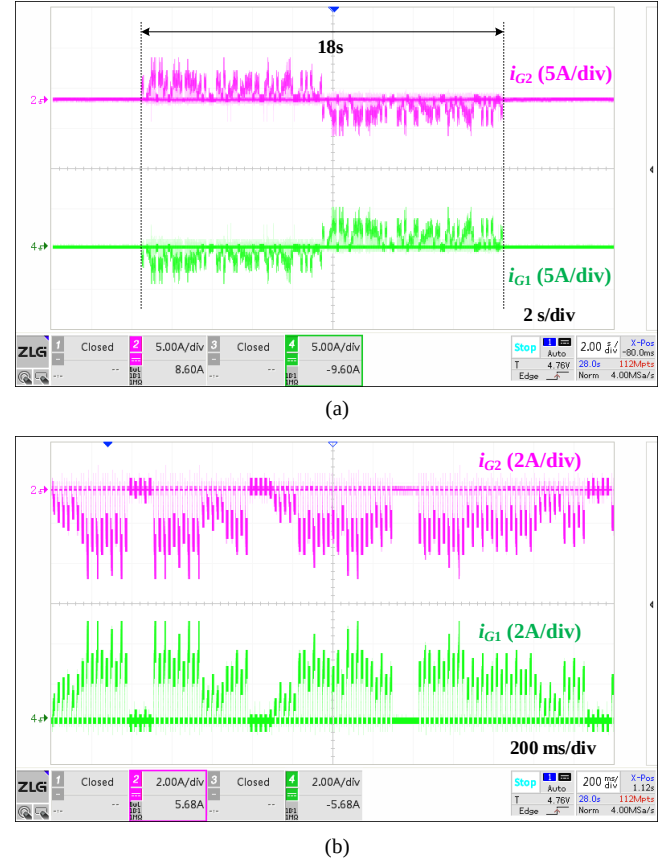


Fig. 20. Measured diagnostic current profiles of the two battery groups. (a) Horizontal resolution is 2 s/div. (b) Horizontal resolution is 200 ms/div.

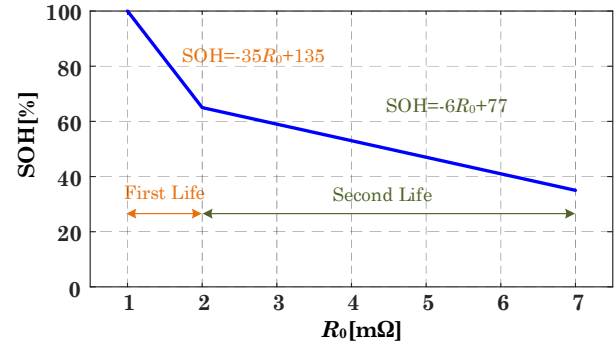


Fig. 21. The relationship between the internal resistance and SOH for a Nissan Leaf battery module [25].

VI. COMPARISON

Table IV consolidates the key metrics of the proposed UBMM and five representative equalizers. The UBMM inherits the autonomous IPOS equalization concept presented in [9], a feature shared only by the AC-bus [27] and the original IPOS configuration. Compared with [9], this paper contributes a multi-module PWM synchronization scheme that enables scalability to 48 UBMM units, as well as an embedded diagnostic circuit with online parameter extraction.

In the switched-capacitor equalizer [4], despite its autonomous operation, the control complexity remains moderate because a large number of high-frequency switches ($2n$) must

TABLE IV
COMPONENTS COUNT AND FEATURE COMPARISON AMONG STATE-OF-THE-ART EQUALIZERS

Feature	Switched-Capacitor [4]	IPOS Module [9]	Flyback [10]	MMB [26]	AC-bus [27]	UBMM (Proposed)
Energy flow type	C2C	M2C	P2C & C2P	HME	M2C	M2C
Modularity	No	Yes	No	Yes	Yes	Yes
Multi-cell equalization	Yes	Yes	No	No	No	Yes
Autonomous equalization	Yes	Yes	No	No	Yes	Yes
Equalization speed*	Slow	Medium	Medium	Fast	Medium	Medium
Control complexity**	Medium	Low	Medium	High	High	Low
Ratio of switch voltage stress w.r.t. cell voltage	1	4	n	4	2	2
High frequency switch count	$2n$	$\frac{n}{4}$	8	$2n + \frac{n}{2}$	$\frac{n}{2}$	$\frac{n}{2}$
Selection switches & drivers	0	0	$4n$	0	0	0
Diodes	0	n	4	0	n	n
Capacitors	$n(n-1)/2$	$2n + \frac{n}{4}$	0	n	n	$2n + \frac{n}{2}$
Inductors	0	$2n + \frac{n}{4}$	0	$2n$	n	$2n + \frac{n}{2}$
Transformer / Coupled inductor	0	$\frac{n}{4}$	2	1	$n/2$	$\frac{n}{2}$
Efficiency (%)***	92.89	86	87	74 / 93	86	86
Number of Tested cells	8	8	12	4	4	16
SOH diagnostic	No	No	No	Yes	No	Yes
Scalability difficulty	quadratic pulse scheduling & sync.	PWM sync. over AC-link unresolved	selection network complexity ↑; speed ↓	multi-winding transformer design limitation	Transformer design bottleneck at scale	Achieved

* *Slow: Non-constant current and non-multi-cell equalization; Medium: constant current or multi-cell equalization; Fast: constant current and multi-cell equalization.*
 ** *Low: switch count $\leq$ cells & autonomous; Medium: switch count $<$ cells or autonomous; High: switch count $>$ cells & non-autonomous.*
 *** *Power dissipation of the selection switches not mentioned.*

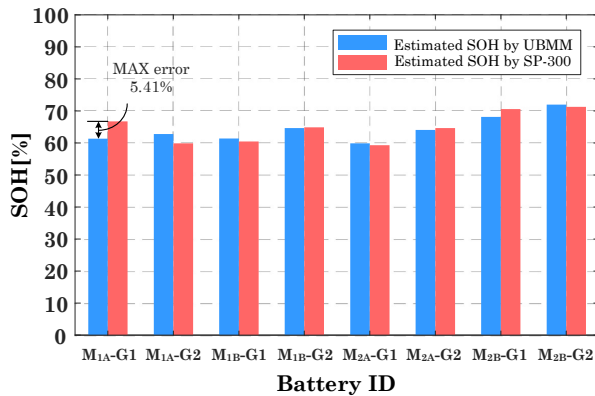


Fig. 22. Comparison of SOH estimated by UBMM and SP-300.

be coordinated through precise pulse scheduling. The flyback-based topology [10] employs a switch network combined with a shared DC/DC converter; consequently, it cannot equalize multiple cells simultaneously, and the equalization speed degrades significantly as the system scales up. The modular multilayer balancer (MMB) architecture [26] adopts a hierarchical modular equalization (HME) scheme that supports energy transfer both within individual modules and across multiple modules. However, because the topology still relies on a shared converter structure, it is inherently limited to processing one module at a time, making multi-cell equalization challenging. Moreover, the design requires a large number

of high-frequency switches, causing both control complexity and implementation cost to escalate rapidly with increased cell count, which poses a significant obstacle to large-scale deployment. Hasanpour et al. [27] proposed a module-to-cell (M2C) architecture that eliminates selection switches and realizes autonomous multi-cell energy equalization. Nevertheless, the system necessitates a substantial number of transformers, which notably raises the implementation complexity.

In contrast, the proposed UBMM circumvents these limitations through a capacitive-coupled AC-link structure, achieves low control complexity, and guarantees scalable multi-module operation via the dedicated PWM synchronization scheme. Collectively, paired with its inherent autonomous operation and embedded diagnostic capability, this design positions UBMM as a highly practical solution for large-scale battery management systems.

VII. DISCUSSION

Regarding the active equalization hardware, the system adopts a capacitively coupled IPOS topology. This structure optimizes the overall hardware cost. Specifically, for a system managed by m UBMMs, the equalization stage strictly requires only m active switches. This configuration eliminates the need for selection switches or multi-winding transformers. Furthermore, the control overhead is minimized. Each module relies on a single switching device operating at a fixed duty cycle and a constant switching frequency. The charging and discharging currents are autonomously distributed based on

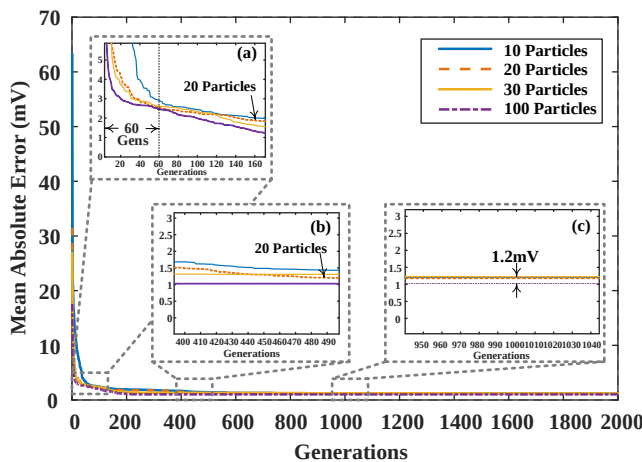


Fig. 23. Convergence profiles of the voltage estimation mean absolute error (MAE) utilizing different m -PSO swarm configurations (10, 20, 30, and 100 particles) over 2000 generations. (a) Early convergence stage demonstrating premature convergence. (b) Mid-stage convergence (400–500 generations). (c) The steady-state stability achieved at 1000 generations.

the voltage differentials across the ac-link, thereby avoiding complex target-group selection logic or closed-loop current control algorithms.

To maintain safe magnetic reset, the effective duty cycle must satisfy $1 - (D + T_{Delay}/T_s) - D_2 > 0$. Under the worst-case voltage scenario ($V_{module} = 7.2$ V, $V_{GM.1.A.2} = 3.2$ V), the theoretical maximum delay is 125 ns. The hardware prototype controls this delay to 30 ns via high-speed differential amplifiers, satisfying the theoretical requirement. To prevent overcurrent, the firmware constrains the maximum duty cycle, and equalization is activated only when the voltage difference reaches 200 mV. Additionally, a 10 A fuse is installed in series with each cell. Efficiency decreases at small voltage gaps. To avoid low-efficiency operation, active equalization stops when the voltage difference falls below 50 mV.

Regarding the embedded battery diagnostic circuit, the hardware implementation relies on a fundamental buck-boost converter topology. This architectural choice simplifies the control logic and minimizes the required active power components compared to multi-stage diagnostic topologies. Within the proposed scalable framework, each UBMM is dedicated to managing exactly two battery cell groups. Consequently, the diagnostic sensing overhead is strictly limited to two voltage sensors and two current sensors per UBMM. The time-averaged power loss is negligible due to intermittent operation, imposing no obvious thermal or efficiency penalty on the system. The maximum pulse current is limited to 4 A, which is far below the rated continuous discharge current of a typical Nissan Leaf battery. The 18 s diagnostic process creates a strictly zero-mean symmetric current profile. The absolute capacity throughput for a single diagnostic event is merely 0.025 Ah, ensuring that the additional cycling stress is profoundly negligible and executable without inducing system downtime.

Configured with 20 particles and 1000 generations, the m -PSO algorithm runs in the background on an STM32F407 microcontroller. As illustrated in Fig. 23, this configuration

achieves an optimal trade-off. Further increases in swarm size beyond 20 particles or iteration count above 1000 generations produce negligible improvements in mean absolute error. The total RAM footprint remains below 50 kB, well within the 192 kB on-chip memory limit. The compiled firmware occupies approximately 50.8 kB of Flash memory, accounting for roughly 5% of the total 1 MB capacity.

During the short diagnostic pulse, the voltage change caused by the slow dynamics is very small. Extracting time constants requires curve-fitting this exponential voltage response, making it sensitive to the random measurement noise in a single diagnostic cycle. Consequently, the algorithm may reach the search boundary for the slow dynamics. This behavior does not affect the extraction of R_0 , which relies on instantaneous voltage edges. Under a constant string current, the DC offset is canceled because the resistance extraction uses current differentials. Under a varying string current, the load-induced voltage superimposes on the diagnostic response and corrupts the identification. For this reason, diagnostics are scheduled during stable current windows. Decoupling methods for dynamic loads are left for future work.

Finally, applying the diagnostic architecture to other chemistries requires offline recalibration of the R_0 -SOH baseline and the OCV-SOC curve. Within the 10 – 30°C window, uncompensated temperature-induced resistance variation (~ 0.1 m Ω) can cause an SOH error of up to 3.5% in the first-life region. Combined with the worst-case single-shot R_0 error (contributing $\sim 5.4\%$ in the second-life region), the overall SOH uncertainty is bounded at $\pm 8.9\%$. In practice, this uncertainty can be reduced by averaging multiple diagnostic cycles and adopting a temperature-dependent lookup table for resistance compensation.

VIII. CONCLUSION

This paper has presented a novel unified battery management module (UBMM) that enables autonomous energy routing, redistribution, and online diagnostics for large-scale battery energy storage systems. The proposed topology leverages a high-frequency capacitively coupled AC link to interconnect all energy paths, allowing for fully autonomous equalization. The circuit achieves adaptive current distribution by battery group voltage deviations without requiring complex control strategies and selection switches. Furthermore, the proposed diagnostic circuit employs a switched-inductor configuration to precisely shape charge/discharge current profiles, facilitating accurate extraction of intrinsic battery parameters via a m -PSO algorithm. The extracted ohmic resistance is directly mapped to SOH, providing a simple yet effective monitoring solution. The UBMM has been experimentally verified in the laboratory using two Nissan Leaf battery modules and has been successfully deployed in a large-scale energy storage system comprising 24 modules, 192 cells, and approximately 400 V.

IX. APPENDIX

To ensure the bidirectional buck-boost converter operates in discontinuous current mode (DCM), the inductor current must

drop to zero within each switching period $T_{sw} = 1/f_{sw}$. Thus, the sum of the inductor charging time (t_{on}) and discharging time (t_{off}) must satisfy $t_{on} + t_{off} \leq 1/f_{sw}$. Based on Faraday's law of induction ($v = L \frac{di}{dt}$), this constraint can be expressed as:

$$\frac{LI_{L-pk}}{v_1} + \frac{LI_{L-pk}}{v_2} \leq \frac{1}{f_{sw}} \quad (A.1)$$

where I_{L-pk} is the peak inductor current, and v_1 and v_2 are the port voltages interfacing with battery group 1 and battery group 2, respectively. In DCM, the input current forms a triangular pulse during the conduction interval. By calculating the area of this pulse over the entire switching period, the maximum average excitation current I_{max} is geometrically related to the peak current and the diagnostic duty cycle D_{diag} :

$$I_{max} = \frac{1}{2} D_{diag} I_{L-pk} \quad (A.2)$$

The peak inductor current can be expressed as:

$$I_{L-pk} = \frac{2I_{max}}{D_{diag}} \quad (A.3)$$

Additionally, based on the inductor charging slope over the on-time interval ($t_{on} = D_{diag}/f_{sw}$), the peak current can be defined as:

$$I_{L-pk} = \frac{v_1 D_{diag}}{L f_{sw}} \quad (A.4)$$

By equating the two expressions for I_{L-pk} , the diagnostic duty cycle D_{diag} can be derived as:

$$D_{diag} = \sqrt{\frac{2LI_{max}f_{sw}}{v_1}} \quad (A.5)$$

Substituting (A.3) and (A.5) back into the initial DCM constraint (A.1) yields:

$$\frac{2LI_{max}}{\sqrt{\frac{2LI_{max}f_{sw}}{v_1}}} \left(\frac{v_1 + v_2}{v_1 v_2} \right) \leq \frac{1}{f_{sw}} \quad (A.6)$$

By squaring both sides and simplifying the algebraic terms, the upper boundary for the critical inductance L is obtained:

$$L \leq \frac{v_1 v_2^2}{2I_{max} f_{sw} (v_1 + v_2)^2} \quad (A.7)$$

In practice, the voltage drops across the interconnection cables are negligible. Therefore, the converter port voltages are equal to the actual battery group terminal voltages ($v_1 = v_{G1}$ and $v_2 = v_{G2}$). Applying this approximation and rearranging the inequality to emphasize the switching frequency constraint yields the final design criterion

$$f_{sw} \leq \frac{v_{G1} v_{G2}^2}{2I_{max} L (v_{G1} + v_{G2})^2} \quad (A.8)$$

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